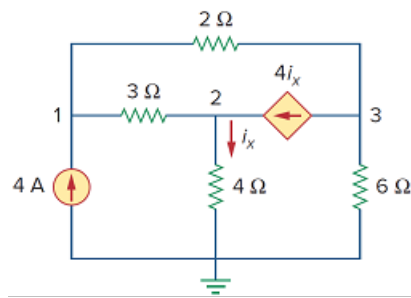


Problem

Find the voltages at the three nonreference nodes in the circuit of Fig. 3.6.

Figure 3.6:



Step-by-step solution

Step 1 of 8

Draw the circuit diagram.

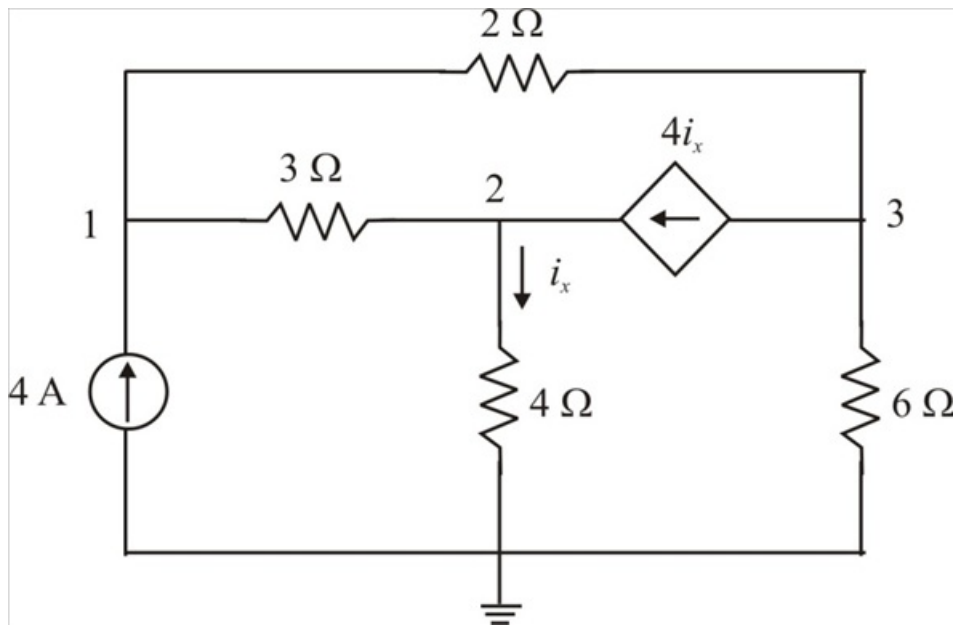


Figure 1

Step 2 of 8

The circuit is labeled as shown in Figure 2.

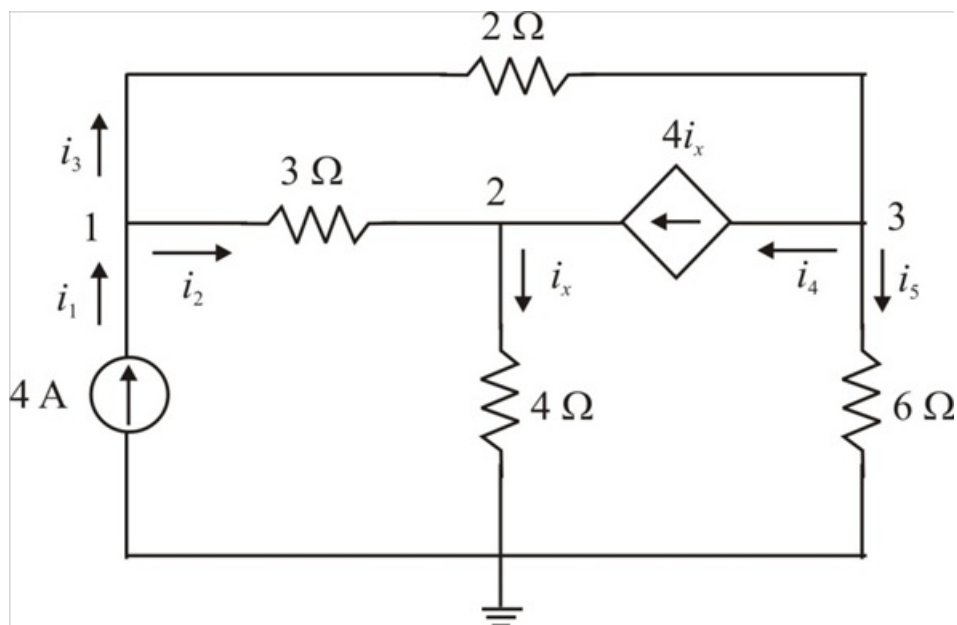


Figure 2

Step 3 of 8

At node 1, applying Kirchhoff's current law, the sum of currents entering any node is equal to the sum of currents leaving from that node.

$$i_1 = i_2 + i_3$$

$$4 = \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{2}$$

Multiplying each term in the last equation by 6, we obtain

$$4 = \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{2}$$

$$24 = 2v_1 - 2v_2 + 3v_1 - 3v_3$$

$$5v_1 - 2v_2 - 3v_3 = 24 \dots\dots (1)$$

Step 4 of 8

At node 2, applying Kirchhoff's current law, the sum of currents entering any node is equal to the sum of currents leaving from that node.

$$i_2 + i_4 = i_x$$

$$\frac{v_1 - v_2}{3} + 4i_x = i_x$$

$$\frac{v_1 - v_2}{3} + 3i_x = 0$$

$$\frac{v_1 - v_2}{3} + 3\frac{v_2}{4} = 0$$

Multiplying each term by 12, we obtain

$$\frac{v_1 - v_2}{3} + 3\frac{v_2}{4} = 0$$

$$4v_1 - 4v_2 + 9v_2 = 0$$

$$4v_1 + 5v_2 = 0$$

$$v_1 = \left(\frac{-5}{4}\right)v_2 \dots\dots (2)$$

Step 5 of 8

At node 3, applying Kirchhoff's current law, the sum of currents entering any node is equal to the sum of currents leaving from that node.

$$i_3 = i_4 + i_5$$

$$\frac{v_1 - v_3}{2} = 4i_x + \frac{v_3 - 0}{6}$$

$$\frac{v_1 - v_3}{2} = 4 \frac{v_2}{4} + \frac{v_3}{6}$$

Multiplying each term by 6, we obtain

$$\frac{v_1 - v_3}{2} = 4 \frac{v_2}{4} + \frac{v_3}{6}$$

$$3v_1 - 3v_3 = 6v_2 + v_3$$

$$3v_1 - 6v_2 - 4v_3 = 0 \dots\dots (3)$$

Substitute equation (2) in equation (1).

$$5\left(\frac{-5}{4}\right)v_2 - 2v_2 - 3v_3 = 24$$

$$-25v_2 - 8v_2 - 12v_3 = 96$$

$$-33v_2 - 12v_3 = 96 \dots\dots (4)$$

Step 6 of 8

Substitute equation (2) in equation (3).

$$3\left(\frac{-5}{4}\right)v_2 - 6v_2 - 4v_3 = 0$$

$$-15v_2 - 24v_2 - 16v_3 = 0$$

$$-39v_2 - 16v_3 = 0$$

$$v_2 = \left(\frac{-16}{39}\right)v_3 \dots\dots (5)$$

Now, substitute equation (5) in equation (4).

$$-33\left(\frac{-16}{39}\right)v_3 - 12v_3 = 96$$

$$13.54v_3 - 12v_3 = 96$$

$$v_3 = \frac{96}{1.54}$$

$$= 62.4 \text{ V}$$

Therefore the node voltage v_3 is 62.4 V.

Step 7 of 8

Next, substitute 62.4 V for v_3 in equation (5).

$$v_2 = \left(\frac{-16}{39}\right)v_3$$

$$= \left(\frac{-16}{39}\right)(62.4)$$

$$= -25.6 \text{ V}$$

Therefore the node voltage v_2 is -25.6 V.

Step 8 of 8

Finally, substitute -25.6 V for v_2 in equation (2).

$$\begin{aligned}v_1 &= \left(\frac{-5}{4}\right)v_2 \\&= \left(\frac{-5}{4}\right)(-25.6) \\&= 32\text{ V}\end{aligned}$$

Therefore the node voltage v_1 is $\boxed{32\text{ V}}$.